

MSC-Trans: A Multi-Feature-Fusion Network With Encoding Structure for Student Engagement Detecting

Nan Xie , Zhengxu Li , Haipeng Lu, Wei Pang, Jiayin Song, and Beier Lu

Abstract—Classroom engagement is a critical factor for evaluating students’ learning outcomes and teachers’ instructional strategies. Traditional methods for detecting classroom engagement, such as coding and questionnaires, are often limited by delays, subjectivity, and external interference. While some neural network models have been proposed to detect engagement using video data, they generally rely on fixed feature combinations, which fail to capture the logical connections and temporal dynamics of engagement. To address these challenges, this article introduces the MSC-Trans Engagement Detecting Network, a temporal multimodal data fusion framework that integrates a convolutional neural network (CNN) and a multilayer encoder–decoder structure. The proposed network includes two key components: first, a multilabel classifier based on ResNet and Transformer, which embeds labels into image features extracted by the CNN for high-precision classification through background inference, second, a temporal feature fusion module, which leverages an encoder–decoder structure to integrate multimodal features over time, enabling stable tracking of classroom engagement. Meanwhile, this open framework allows users to freely select feature combinations for temporal fusion based on specific scenarios and needs. The MSC-Trans Engagement Detecting Network was validated on the DAiSEE dataset, augmented with real classroom data. Experimental results demonstrate that the proposed method achieves state-of-the-art performance in continuous engagement tracking metrics, with flexible and scalable feature selection. This work offers a robust and effective approach for advancing engagement detection in educational settings.

Index Terms—Multilabel classification, student engagement, temporal feature fusion.

I. INTRODUCTION

LEARNER engagement has always been a significant area of research for educational researchers. There is a close

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relationship between students’ learning outcomes and their level of engagement. Students who are highly engaged and actively participate in classroom interactions are more likely to absorb new knowledge and concepts, deepen their understanding of the teaching topics, and reduce distractions that lead to lack of concentration and missing key teaching content. Therefore, real-time analysis of learner engagement has a positive effect on improving teachers’ teaching strategies and promoting active learning, which has triggered the demand for high-precision tracking of learner engagement in real time.

However, it should be noted that current methods of monitoring engagement primarily rely on coding, questionnaires, and other similar approaches, with intelligent methods based on machine learning rarely considering the various factors influencing engagement and their evolution over time. As the integration of information technology and education deepens with the further development of intelligence in the education sector, emerging technologies, such as multimodal sequence encoding networks and deep learning methods are promoting the formation of a digital-driven educational model. The education sector can use more tools to analyze students’ behaviors and engagement levels. These technologies can be integrated into teaching and testing phases through multimedia teaching equipment and networks, allowing for real-time tracking of learner engagement in video form. This is expected to improve teaching and learning processes, offering new perspectives for researching class participants’ engagement while also presenting new challenges.

A. Theories on Engagement Detecting

With the informatization of society, information technology education has evolved beyond mere computer technology training to an education that adapts to the societal demands for information, aiming to train students’ information literacy. To enhance class efficiency and improve teaching outcomes in interactive modes, the construction of engagement theory and methods of monitoring engagement are particularly important. To analyze the current state of research on classroom participant engagement detection, it is necessary to identify the existing gaps and directions for improvement by summarizing related research from two aspects: the theoretical foundation of engagement and the development of engagement detection technologies.

Classroom teaching, as a critical component of student education, serves as the main place where students receive knowledge, engage in practice, and develop their thinking abilities. The level of student participation in the classroom is also an important indicator for assessing the quality of education. This state of classroom participation, also known as “engagement,” is considered by Mosenthal [1] as classroom involvement based on the learner’s cognitive and emotional systems. Skinner and Wellborn’s [2] research also indicates that “engagement” is a description of a multidimensional learning phenomenon, encompassing the behaviors and emotional expressions of learners, which includes proactive behavior, effort, and persistence in the classroom environment, as well as the emotional state of the learner during learning activities.

Researching students’ classroom engagement can help the educator to analyze teaching outcomes from the perspective of the classroom, as engagement reflects students’ interest in learning, which significantly impacts their grades [33]. Specifically, students with high levels of classroom engagement are superior learners; their highly focused state in the classroom can amplify intrinsic learning motivation, suppress other motivations, and reinforce self-regulation strategies to help them grasp deeper conceptual knowledge [34].

To analyze students’ classroom engagement, D’Mello et al. [3] first described students’ engagement through some observable features, including measurable characteristics, such as changes in facial expressions and classroom movements. As mentioned earlier, engagement is a description of a multidimensional phenomenon. Therefore, Azevedo [4] and other researchers suggested that when analyzing engagement, multiple sources of classroom data should be integrated. This approach correlates learners’ classroom states with processes related to behavior, cognition, motivation, emotion, and metacognitive engagement, thereby analyzing and inferring students’ self-perceptions in terms of emotions, knowledge, and values before, during, and after learning.

A typical class lasts for about 40 min. In such a long duration of teaching activity, there is an interdependent and lagging relationship among student engagement, intrinsic learning motivation, and the length of learning time. This indicates that student engagement is not constant but changes over time due to various factors. Therefore, student engagement can be predicted based on time points through their intrinsic motivation.

Some case studies have shown methods for enhancing student engagement in different contexts. Virtanen and Lerkkanen [35] examined the relationship between three dimensions of classroom quality (emotional, organizational, and instructional support) and behavioral engagement in Finnish secondary schools using the Classroom Assessment Scoring System. Gong and Pang [36] investigated the challenges and strategies of engaging students in online Chinese as a Foreign Language (CFL) classrooms through teacher interviews and observations, proposing strategies such as leveraging multilingual resources, technology-enhanced feedback, and personalized interactions. Fernandez-Terol and Domingo [37] focused on a school’s transition from traditional to active classrooms via project-based learning, identifying teachers’ challenges, needs, and the

organizational support required for this methodological shift. Collectively, these studies highlight the critical role of organizational and instructional support, emotional engagement, and innovative teaching methodologies in fostering student engagement.

The research findings mentioned above indicate that engagement can reflect students’ educational situations from the perspective of classroom teaching. Educational researchers can measure and assess engagement from multiple dimensions, and the changes in engagement have characteristics that develop over time, providing a strong reference for teachers’ instruction.

B. Main Contributions

- 1) MSC-Trans Engagement Detecting Network, a temporal feature fusion network with a two-stage Transformer structure, is proposed, which enables the classification of student classroom engagement features through video, and tracks and predicts engagement through the scalable temporal feature fusion. This network can provide feedback on student engagement levels in real classroom settings, enabling teachers to better understand students’ engagement status, adjust their teaching strategies, optimize their teaching methods, and improve classroom quality and student learning outcomes.
- 2) At the step of multilabel classification, a C-Tran-based multilabel image classification network with a logical reasoning structure was applied. By using mask encoding for features and background elements, this approach better classifies student classroom image features while also determining the occurrence intensity of each feature category.
- 3) At the step of multimodal feature fusion phase, the multi-head self-attention encoding structure of the Transformer is used to construct scalable combinations of sequential features. This enables the tracking and prediction of temporal dynamics in freely personalized modality combinations.

C. Structure of the Article

The rest of the article is organized as follows. Section II introduces related works in engagement detection. Section III presents our multilabel temporal feature fusion engagement detection network. Section IV introduces the experimental process and proves the validation of our detection method with a case study in real class. Section V summarizes this study and looks forward to future work.

II. RELATED WORKS

Engagement has multidimensional descriptive characteristics; therefore, when monitoring the engagement status of classroom participants, researchers have developed the following methods for monitoring the engagement status of classroom participants, considering factors such as the learning context and research objectives:

A. Methods Based on Questionnaires, Observation, and Behavior Coding

Educational psychology categorizes student classroom engagement into four dimensions: skill engagement, teacher–student interaction, emotional engagement, and performance engagement [5]. Therefore, by designing structured dimensions of questions, levels of questions, and question language, it is possible to construct questionnaires and expert observation systems based on these four dimensions. Combined with student participation self-reports, learning goal setting, and students' academic performance, student engagement can be analyzed and researched [6].

Questionnaire systems can collect standardized feedback from students and teachers on a large scale and analyze engagement from the intrinsic cognitive perspective of classroom participants. However, these self-reported data are subjective, and their accuracy can be influenced by the social expectations and recall biases of the classroom participants. In addition, questionnaire systems have a latency issue, unable to monitor classroom engagement states in real time.

Systems based on observation and coding can achieve real-time tracking of engagement states and provide quantifiable multidimensional data. However, they require significant time and human resources and may disrupt the normal proceedings of the classroom.

B. Methods Based on Machine Learning

To reduce the subjectivity in questionnaire and observation methods, machine learning-based methods can be used to analyze and classify classroom participants' behavior coding [7]. By vector encoding student behavior records, cognitive states, and facial emotions, machine learning methods can predict the level of students' classroom engagement and provide scaffolding for classifying learner engagement from cognitive and behavioral congruence dimensions [8], [9].

Machine learning-based methods can classify learners' classroom states based on a large amount of data, improving the objectivity of engagement recognition to some extent, and also achieving real-time monitoring of engagement. However, there are still shortcomings in complex pattern recognition, model transfer capabilities, and classification of multimodal data.

1) *Engagement Detection Based on Computer Vision:* With the development of image recognition technology based on convolutional neural networks (CNN), neural networks have become a common means for feature extraction and classification of facial expressions. As a result, researchers have begun to explore the use of image-based neural networks to recognize learners' facial expressions to monitor engagement. Pabba and Kumar [10] chose to first use MTCNN to separate facial expressions negatively correlated with attention states, such as “boredom,” “confusion,” “focus,” “frustration,” “yawning,” and “sleepiness” from collected learner facial images. Then, a secondary CNN is constructed to train on these negative emotional facial features, enabling early warning of large-scale negative engagement states in student groups. The trained model has achieved train and test accuracy of 78.70% and 76.90%, and the proposed methodology gave promising results when

compared with self-reported engagement levels by students. Pise et al. [11] noted the continuity between learners' expressions and used a temporal relation network (TRN) to recognize the emotional changes in students' faces during online learning processes. TRN performs sparse sampling on single frames and then learns the causal relationships between them, considering both single-scale and multiscale temporal relationships, to achieve long-term tracking of student engagement. Furthermore, the feasibility of using multilayer perceptron as a baseline classifier framework for end-to-end training in video-based facial emotion recognition was also tested.

In the process of engagement recognition using image neural networks, researchers such as Mandia et al. [12] realized the need to construct standard engagement training-testing datasets to improve the confidence and universality of network predictions. Based on existing COCO and VOC image recognition task datasets, Gupta et al. [13] created the DAiSEE real-scene student engagement video dataset by massively collecting facial video data from students in online university classes. This dataset includes 9068 video segments from 112 participating students, categorized into four levels of engagement: very low, low, medium, and high.

2) *Engagement Detection Based on Temporal Series:* Engagement recognition networks based on single images overlook the developmental nature of learners' engagement over time and have a weaker ability to characterize the dynamic properties of learners' intrinsic motivation. To address this issue, researchers have explored how to use neural networks with temporal characteristics to monitor learners' engagement.

Long short-term memory (LSTM) recurrent neural networks (RNNs) enhance RNNs with the addition of “memory gates” and “forget gates,” enabling dynamic processing of long sequence information. Therefore, Zhou and Wu [14] utilized the LSTM network to improve upon image neural networks, proposing a multimodal long short-term deep regression network based on facial expressions, upper limb movements, and environmental features. By integrating the long-term and short-term dependence features of student images, this network can accurately predict students' classroom engagement over extended monitoring periods.

LSTM networks can dynamically optimize the content of long-term memories and what to forget, but the complexity of their underlying RNN networks also leads to issues, such as high computational cost and susceptibility to overfitting. Therefore, researchers such as Du et al. [15] have turned their attention to temporal convolutional networks (TCN), a type of one-dimensional causal convolutional network based on CNNs that can extract features from sequence information over time steps in a convolutional manner. They summarized the latest developments and cutting-edge technologies in learner engagement detection methods based on TCN and Resnet networks. Abedi and Khan [16] leveraged the cross-temporal feature extraction properties of TCN to expand the problem of student engagement detection into a spatial-temporal classification problem occurring in both space and time, proposing a hybrid neural network structure that combines residual networks with TCN. In this structure, two-dimensional residual networks extract spatial features from continuous video frames, while TCN analyze the

temporal changes in video frames to detect the level of student engagement.

The application of TCN in engagement recognition introduces the concept of serializing classroom participants' learning behaviors. However, its characteristic of one-dimensional convolution also poses challenges to the multimodal recognition of learners' engagement, which is not conducive to a comprehensive assessment of classroom states. Moreover, both TCN and LSTM utilize a sliding window approach to process temporal information according to causal logic, which can lead to the loss of logical connections between windows when modeling information over longer periods.

3) *Engagement Detection Based on Multimodal Fusion:* Learner engagement states can be characterized not only by emotions reflected in facial expressions but also include various behavioral and cognitive dimensions. Building on the foundation of recognizing engagement through facial images, researchers such as Ayoub et al. [17] have attempted to incorporate representations of engagement from other modalities. Altuwairqi et al. [18], against the backdrop of online classes, proposed a multimodal fusion neural network that combines facial expressions, keyboard inputs, and mouse movement data. After comparing methods using single data forms, bimodal forms, and multimodal forms, it was found that the multimodal approach achieved the highest accuracy in detecting student classroom engagement. Sümer et al. [19] opted to combine facial expressions with learners' head postures, proposing a combined neural network based on students' facial expressions and head postures, and embedded a multidimensional channel for emotional features and attention, achieving recognition of student classroom engagement in real environments. A time-dependent mechanism for engagement was also proposed, with network recognition over a 60-s span based on time series outperforming recognition based on a single image. In addition, researchers such as Chaouachi and Frasson [20] have monitored engagement through brainwave signals, leading Xiao and Jiang [21] to propose a multimodal fusion neural network based on brainwaves, eye movements, and facial expression data, and conducted comparative tests to explore the effects of multimodal combinations on engagement recognition. It was concluded that the combination of eye movements and facial expressions was most effective in detecting engagement in learning behaviors; the combination of brainwaves, eye movements, and facial expressions was most effective in detecting cognitive and emotional engagement.

a) *Methods of multilabel classification:* The teaching environment's impact on students and the interaction between teachers and students can influence engagement significantly. Therefore, on the basis of Andreas' [22] research on environmental emotional detection, Savchenko et al. [23] incorporated the teacher's focus state, emotional state, and teacher-student interaction as dimensions reflecting classroom participants' engagement states. A complex engagement recognition network is designed based on facial expressions, combining individual and group emotions of students and interactions between students and teachers. This network predicts the overall emotional atmosphere of the classroom participation group and provides real-time predictions of individual student engagement.

b) *Methods of temporal series fusion:* Researchers propose using the Transformer encoder structure for modeling and analyzing temporal, multimodal behaviors of classroom participants. The Transformer is an encoder and decoder network structure based on self-attention mechanisms. Unlike TCN and LSTM, the Transformer does not require a sliding window and can process the entire sequence in parallel. Its unique multihead self-attention mechanism allows the network model to focus on information at any position in the input sequence from other positions, capturing hierarchical relationships and logical connections between different dimensions and levels of information. This capability enables it to competently handle both temporal sequence modeling and multimodal modeling tasks in engagement monitoring.

Guo et al. [24] improved an encoder network based on the V-Transformer, enabling it to detect changes in student classroom engagement in the spatiotemporal fusion dimension with high performance and low computational complexity. Combining and improving existing CNN networks with the Transformer is a common approach among researchers. Xiong et al. [25] used a combination of CNN and Transformer in a multimodal neural network to merge large-scale student movements with small-scale student facial expressions, suppressing the impact of background noise from multiple people on student engagement recognition and enhancing the recognition rate of learning engagement in real teaching scenarios. Huang et al. [26] further combined the Transformer with TCN and LSTM to propose a composite recognition network for student classroom engagement, which has high accuracy and stability over long time spans. DU [15] combined the encoder structure of the Transformer network with one-dimensional and two-dimensional fully convolutional networks to form a spatiotemporal hourglass CNN. By integrating low-level encoding and high-level decoding cues to extract contextual relationships, the network is guided to gradually predict the macro long-term trends and refined short-term fluctuations of learner engagement.

Through the study of multilabel prediction methods and sequence fusion methods, we found that when detecting and tracking students' engagement based on visual data, using multilabel prediction methods can integrate various key features reflecting student engagement and fuse computations, improving the efficiency of video information utilization and detection accuracy. Meanwhile, sequence fusion methods can give the mapping relationship between student behavior and engagement temporal continuity, offering better real-time performance and consistency in tracking student engagement. Therefore, combining multilabel prediction methods with sequence fusion methods for predicting and tracking student engagement is a promising approach worth researching.

III. MSC-TRANS ENGAGEMENT DETECTING NETWORK

This research takes student learning behaviors and expressions as the basis for engagement reference, using facial expressions (eye state), head posture (whether facing forward), writing status, and off-task behaviors as feature indicators. It proposes a method for tracking and measuring engagement through the

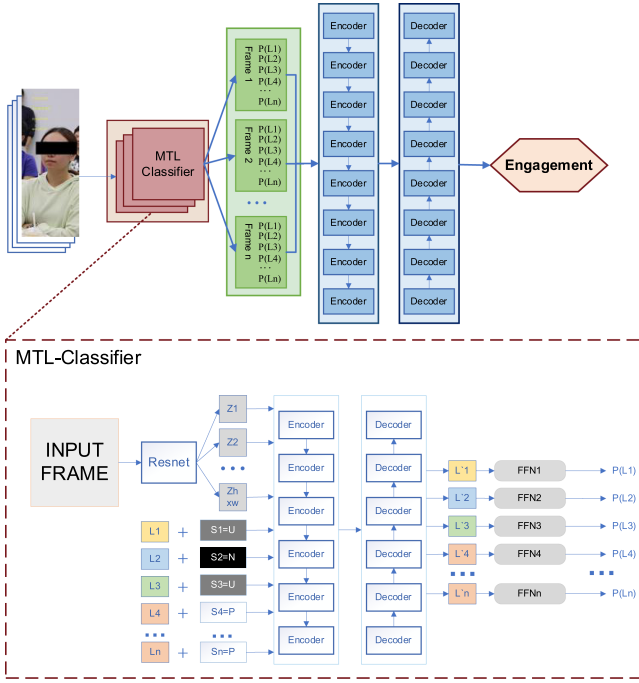


Fig. 1. MSC-trans engagement detecting network structure.

fusion of temporal multimodal data, which combines CNNs with multilayer encoder networks.

A. Main Network Structure

The method consists of an image feature extraction module, an encoder logic reasoning classification module, and an encoder temporal feature fusion module. The image feature extraction module extracts student classroom video image features in chronological order through the Resnet network [27].

The encoder logic reasoning module classifies the temporally arranged image features logically based on the logical relationships between feature labels (for example, when both eye and head states are negative, it is more logical to infer that the writing state is also negative).

The encoder temporal feature fusion module fuses the classified image features in chronological order for inferencing, obtaining temporal engagement prediction results. The entire network is shown in Fig. 1.

B. Multilabel Classifier

Multilabel image classification is the task of predicting a set of labels that correspond to objects, attributes, or other entities in an image. In classifying student classroom imagery described by multiple labels, we adopted a general framework for multilabel image classification that utilizes the Transformer encoder structure [28] to handle complex dependencies between visual features and labels. Our method consists of a set of Transformer encoder structures and a visual feature extraction network provided by a CNN. It primarily goes through two stages to establish complex relationships between multiple labels and image features.

- 1) Background label inference: In the classroom scene, feature combinations centered around students possess logical connections rather than being isolated from each other. Therefore, the model treats the frequent items in the image labels as the “background” for multilabel inference, and under the weighting of this background label, it combines with state labels to infer and predict the distribution of unknown image labels. Such labels can provide logical support for accurate inference in complex scenarios.
- 2) Partial label inference: Building on the foundation of background label inference, it is often unnecessary for all label combinations to participate in the multilabel prediction of image features. The encoder network, after training, achieves the prediction of a complete set of target labels in the presence of missing labels in the input label group. Such partially missing label embeddings ensure the stability of the network in highly occluded scenarios.

To achieve the goal of enabling the model to obtain background labels and perform partial label inference, three types of embedding features are set up in the classifier structure.

Student Image Feature Embedding Z: Given an input image $x \in \mathbb{R}^{H \times W \times 3}$, the feature extractor outputs a tensor, where h , w , and d represent the output height, width, and depth, respectively. Then, each vector $z_i \in \mathbb{R}^d, i \in [1, P], P = h \times w$ in Z can be considered as representing a subregion corresponding to a block in the original image space.

Behavior Label Embedding L: For each image, a set of label embeddings $L = \{l_1, l_2, \dots, l_\ell\}, l_i \in \mathbb{R}^d$ are extracted, which represent ℓ possible labels in the prediction domain. The method of embedding labels is learned from an embedding layer of size $d \times \ell$.

Additional State Label Embedding S: To set the initial background labels and missing labels, the state embedding feature vectors $s_i \in \mathbb{R}^d$ are added to the given labels in the following manner:

$$\tilde{l}_i = l_i + s_i. \quad (1)$$

The vector s_i has three states: Unknown State U, Negative State N, and Positive State P. State embeddings enable users to 1) avoid using any prior information by adding unknown embeddings, 2) utilize partial labels or additional information by adding negative and positive embeddings to those labels, and 3) test interventions in the model by asking, “How would the prediction change if this label were set to positive (negative)?”

These three types of embedded content enable the encoder network to add additional state information to behavior labels by assigning positive and negative polarizations, automatically setting missing value masks during training, and anticipating behavioral characteristics when labels are missing, thereby enhancing the robustness of the encoder network.

The Transformer network can capture rich dependence information between variables, making it easy to jointly embed image features and behavioral labels. A series of image features embedded together with additional behavioral label features are transmitted to the encoder as $H = \{z_1, \dots, z_{h \times w}, \tilde{l}_1, \dots, \tilde{l}_\ell\}$. The weight calculation of each embedding information h relative to other embedding information $h_j \in H$ is achieved through

self-attention mechanism. To obtain the relative weights between embedding information, first, the standard normalized attention coefficients α_{ij} between embedding information h_i and h_j are calculated as

$$\alpha_{ij} = \text{softmax} \left(\frac{(\mathbf{W}^q h_i)^T (\mathbf{W}^k h_j)}{\sqrt{d}} \right). \quad (2)$$

Then, the weighted sum of each embedding feature information is calculated as

$$\bar{h}_i = \sum_{j=1}^M \alpha_{ij} \mathbf{W}^v h_j. \quad (3)$$

A nonlinear ReLU activation function [see (4)] is applied to update the sum of each weighted embedding information $\bar{h}_i$

$$h'_i = \text{ReLU}(\bar{h}_i \mathbf{W}^r + b_1) \mathbf{W}^o + b_2 \quad (4)$$

where $\mathbf{W}^k$ is the key weight matrix, $\mathbf{W}^q$ is the query weight matrix, $\mathbf{W}^v$ is the value weight matrix, $\mathbf{W}^r$ and $\mathbf{W}^o$ are transformation matrices, b_1 and b_2 are bias vectors, and the learning matrix $\{W^k, W^q, W^v, W^r, W^o\} \in \mathbb{R}^{d \times d}$ is not shared between layers. This process can be iteratively applied in L layers to update h'_i in the label embedding layer, and then input it into the encoder layers, resulting in the final output $H' = \{z'_1, \dots, z'_{h \times w}, l'_1, \dots, l'_\ell\}$ of the encoder after L layers.

In the end of the classifier structure, by using the Transformer encoder to model the dependencies between features and labels, the classifier can predict the final labels.

The whole classifier workflow can be described as follows.

- Step 1:* Input the image and process it initially to extract the initial features in the image.
- Step 2:* Extract deep features through multiple residual blocks.
- Step 3:* Embed behavioral labels, as well as additional state labels in deep features.
- Step 4:* Extract global contextual information of deep features with their embedded labels by the encoder.
- Step 5:* The decoder generates the output sequence step by step and returns the complete prediction result.

C. Coded Temporal Feature Fusion

With the final label information obtained by the classifier, the network transmits the label vector to the Transformer encoder, which is then converted into a set of vector representations. This one-dimensional time-series data transforms the feature labels of each image into a fixed-size vector $(x_1, x_2, \dots, x_n)$ through word embedding, treating each image's feature label as a word or token. Subsequently, the vector is subjected to position encoding to maintain the sequence's order information. At this stage, the length of the input label vector can be defined by the user to meet the requirements for the sequence fusion of different feature combinations.

Once the vectors enter the encoder layer, the Transformer's self-attention mechanism computes the relationships between all pairs of feature vectors in the input sequence. The output produced at this step is a weighted representation $\mathbf{z} = (z_1, z_2, \dots, z_n)$ of the input, where each image feature pays

more attention to other image features associated with it. Subsequently, a feedforward neural network processes the representation of each position independently, based on the output of the self-attention mechanism.

After obtaining the attention-weighted encoding of the image label vectors, the group of feature vectors enters the decoder layer. Here, an encoder-decoder attention mechanism enables the decoder to focus on the encoder's output, generating an output sequence $(y_1, y_2, \dots, y_m)$ based on the input sequence. This is then processed by a feedforward neural network like the one in the encoder for each position's representation.

The decoder's output is transformed into prediction results through a linear layer and a SoftMax layer, obtaining the probability distribution of the categories to which the image feature combinations belong.

D. Data Forms

In an MTL-Classifier structure, pretrained Resnet101 Network on ImageNet is used as the feature extractor; its input image dimension is 640×640 . The images undergo random flipping and random center cropping at the input stage, resulting in processed image sizes of 576×576 . After feature extraction, the output size is a tensor of $18 \times 18 \times d$, thus totaling 324 feature embeddings.

When configuring the encoder structure in the MTL Classifier, to allow a specific embedding to pay attention to multiple other embeddings (or groups), C-Tran utilizes a four-head attention structure and employs a six-layer Transformer block. Between each embedding update block and layer norm, there is a residual layer, optimizing training. Adam [29] is used as the optimizer with $\text{betas} = (0.9, 0.999)$, $\text{weight decay} = 0$, and the network structure is optimized with regularization dropout [30] of $p = 0.1$.

In the temporal feature encoder structure, the encoder and decoder transform the data's representation format through attention mechanisms and feedforward networks, keeping the data dimension (vector size) constant. Meanwhile, the information content of the data tensor is merged and fused through the multihead self-attention stage, linking more contextual image feature information.

IV. EXPERIMENTAL DETAILS

A. Dataset Structure

To verify the feasibility of the model, this study conducts experiments based on the following two datasets.

1) *DAiSEE Dataset:* The DAiSEE (Database for Affect in Speech Emotion) [13] is a multimodal dataset for emotion recognition, focusing primarily on identifying emotions from speech and facial expressions. It includes audio and video data of 113 students from various countries and cultural backgrounds in classroom environments. The videos, ranging from 20 s to 1 min, are segmented based on the engagement state, with annotations for each student's engagement state in the txt format provided for each short video. These data were collected in real classroom settings with the aim of assisting researchers in developing

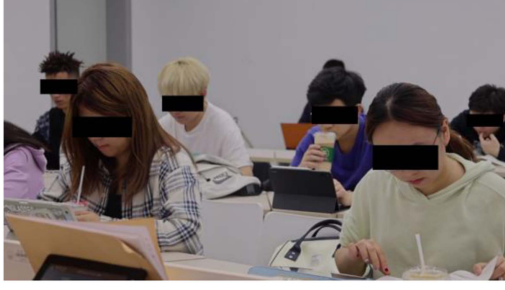


Fig. 2. Offline teaching scenario.

algorithms for automatically recognizing student engagement states. Evaluating students' engagement levels in learning environments can help improve the educational experience.

2) *University Classroom Live Dataset*: Based on the formats of the DAiSEE dataset and the VOC2007 [31] dataset, a local dataset was created using video recordings of students in classroom settings. Initially, videos capturing students' facial expressions and behavioral actions were collected in a university computer network class. Subsequently, teachers and experts rated the students' engagement states based on the classroom footage. This was combined with self-reported engagement states from student questionnaires and classroom quiz results to define the real values of engagement levels in the dataset.

a) *Data collection*: To create a realistic classroom environment dataset, we collected video data from students in a college computer networking class and obtained standard values for engagement scores corresponding to the classroom footage through expert grading, surveys, and tests. These standard values of classroom engagement were derived from teacher reports, student reports, and third-party expert observations, and were matched with timestamps corresponding to student and teacher behavior characteristics.

Cameras were placed at the front sides of the classroom to capture students' facial expressions, body, and head postures. Students were randomly distributed in the classroom, and their distribution characteristics were unrelated to gender or academic performance, as shown in Fig. 2.

b) *Data preprocessing*: To ensure the effectiveness of the experiment, the following facial and behavioral characteristics were selected as the basis for engagement scoring: eye state, head posture, writing status, and mobile phone usage. Specifically, the eye state refers to eyes open wider than 60% as positive samples, open between 30% and 60% as neutral samples, and less than 30% as negative samples; head posture refers to the angle between the facial plane normal and the student's direct front being less than 15° as positive samples, between 15° and 30° as neutral samples, and greater than 30° as negative samples. The distribution of these characteristics in the dataset is given in Table I.

B. Evaluation Indicators

Mean-squared error (MSE) is introduced as the evaluation metric for classifying students' classroom behavior and facial expressions in individual images, and we use a calculation method based on the similarity of the Fréchet Distance [32] to

TABLE I
DATASET DISTRIBUTION

	Positive %	Negative %	Neutral %
Eyes status	60	10	30
Head status	50	20	30
Writing	40	60	0
Using phone	10	90	0

assess the effectiveness of the network in temporal tracking of students' classroom engagement.

A smaller MSE value indicates a better fit of the model to the data, meaning that the difference between the model's predictions and the actual values is smaller. A lower MSE demonstrates that our model has high accuracy in inferring and predicting the state features of students in individual images. During the phase of sequence fusion prediction tasks, a higher Fréchet distance similarity score indicates that the network model has better stability and temporal accuracy in tracking changes in students' engagement levels.

The Fréchet curve similarity is a method used to measure the similarity between two curves. This method is based on the Fréchet distance, which represents the maximum possible value of the minimum continuous movement between two kinematic paths. In the task of engagement tracking curve similarity detection, two sequences of points representing the actual values $A = \{a_1, a_2, \dots, a_m\}$ and the predicted values $B = \{b_1, b_2, \dots, b_n\}$ are given to denote the points on the two curves. A two-dimensional array $dp[i][j]$ is defined to store the minimum Fréchet distance from the first i points of sequence A to the first j points of sequence B. Then, $dp[i][j]$ can be calculated through the following recursive relation:

$$dp[i][j] = \max \left\{ \min \left\{ \begin{array}{l} dp[i-1][j], \\ dp[i-1][j-1], \\ dp[i][j-1] \end{array} \right\}, d(a_i, b_j) \right\} \quad (5)$$

where $d(a_i, b_j)$ is the Euclidean distance between point a_i and point b_j . The initial condition is $dp[0][0] = d(a_1, b_1)$, and for all $i > 0$ and $j > 0$, it is necessary to calculate all $dp[i][j]$ values before computing $dp[i'][j']$, where $i' \leq i, j' \leq j$. Ultimately, $dp[m][n]$ represents the Fréchet distance between the actual value sequence A and the predicted value sequence B, with its magnitude reflecting the stability of the predicted curve in tracking the actual curve.

In the phase of student image feature time series fusion based on the Transformer encoder, the network model's temporal prediction of student engagement will have a delay of a frames, where a represents the length of the sequence segment required for the fusion network inference. Under the effect of the encoder network's self-attention mechanism, the number of frames a covered by each prediction may vary. If the MSE method, which corresponds exactly to time and frame, is used as the metric for sequence tracking prediction effects, it is not feasible to simply shift the sequence by a frame for corresponding calculations. Therefore, the MSE method cannot adequately evaluate the predictive ability of the sequence fusion network under delay

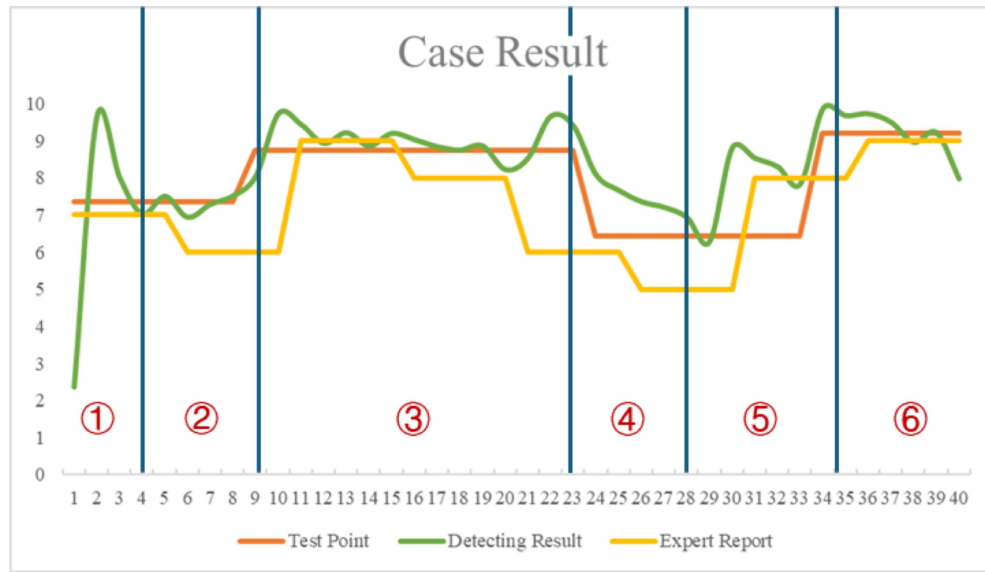


Fig. 3. Case result.

tracking characteristics. Using the Fréchet curve similarity calculation can reflect the overall similarity between the predicted tracking curve and the actual engagement value curve, avoiding the instability of evaluation effects caused by the abrupt changes in MSE due to irregular delays.

C. Experimental Conducting

The DAiSEE dataset is evenly mixed with 80% of the college classroom live dataset. Using OpenCV, one frame is extracted every 1 s, and the extracted frame images are cropped to retain only the visible area of a single student. After image extraction and cropping, the images and corresponding annotation information are rearranged according to the Pascal VOC2007 dataset format and loaded into an MTL Classifier for training, setting the number of training epochs to 100, batch size to 24, and learning rate to 0.01.

The part of the college classroom dataset used for training the MTL Classifier that contains annotations of students' classroom behavior is extracted and arranged in a time series to form a student behavior matrix. This matrix, along with the engagement truth sequences defined by expert surveys at corresponding time nodes, forms a sequential dataset. This dataset is loaded into a temporal feature fusion encoder for training, setting the number of training epochs to 50, base sequence length a to 7, and learning rate to 0.01.

After training the MTL Classifier and the temporal feature fusion encoder, the remaining 20% of the college classroom live dataset is used as a validation set to separately test the student engagement detection performance of the MTL Classifier and the overall performance of the classifier-fusion encoder.

D. Case Study

In order to verify the feasibility of the MSC-Trans network in a real classroom, a 40-min-long computer network lesson was

TABLE II
CORRELATION COEFFICIENTS BETWEEN THREE ELEMENTS

	DR-ER	DR-TP	ER-TP
Correlation Coefficient	0.431	0.552	0.639

selected as a case study. In this experiment, 30 students with similar learning levels were tested, the course covered four main knowledge points in computer networks, and the instructor's teaching methodology consisted of lecture-based teaching and interactive teaching.

During the image preprocessing stage, the system first crops individual student images to minimize the impact of background interference. When student images contain occlusions or missing features, the monitoring network's MTL Classifier combines background and missing-label predictions to achieve feature completion through contextual reasoning. In addition, the self-attention mechanism in the transformer's encoder dynamically focuses on key regions within the input image, effectively reducing the influence of nonessential information in the environment, such as cluttered backgrounds or temporary distractions. This multilayered anti-interference design enhances the system's reliability and applicability in real classroom settings.

The MSC-Trans system tracks student engagement in real time and generates engagement scores ranging from 0 to 10, with higher scores indicating higher classroom attentiveness. As a validation, this study introduces in-class expert observation and postclass knowledge quizzes as research benchmarks. In-class expert observation means that two experts with teaching experience observe students' behavior in real time and assess their concentration based on behavior, demeanor, and other criteria; the after-class knowledge point quiz is for students to complete the quiz questions designed for this section of the course to test their mastery of the four computer network knowledge points,

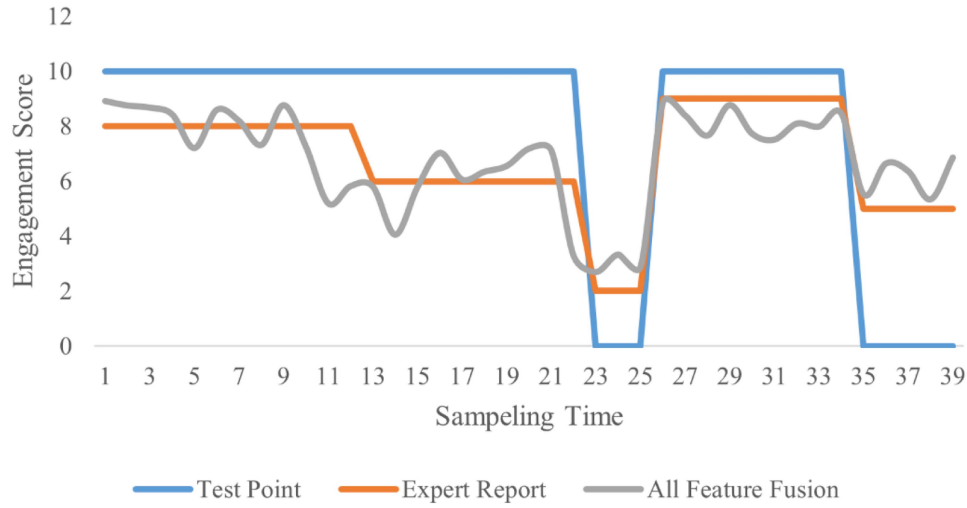


Fig. 4. Experimental result.

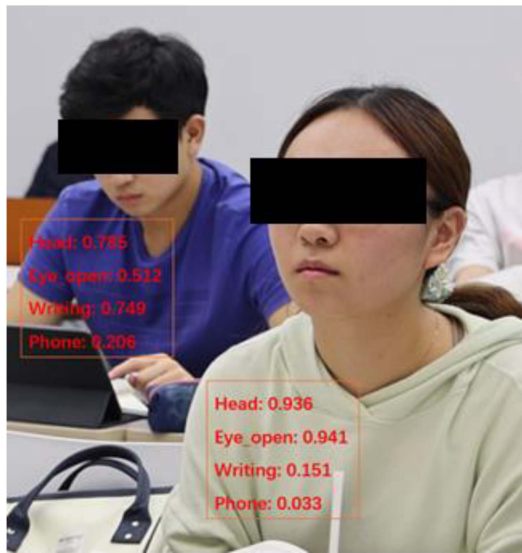


Fig. 5. Result of classification.

and the scores of the four knowledge point tests will be regressed by the correction of the difficulty coefficient.

In the experiment, the detecting system records the average of students' engagement in different knowledge points over time, and its results against the expert-observed and corrected scores on the knowledge point quiz are shown in Fig. 3.

The correlations between the engagement detecting system's result (DR) and the expert observation results (ER), as well as the performance on the postcourse knowledge test points (TP) were calculated separately, and the results are given in Table II.

It can be seen that there is a moderately strong correlation between the results of the MSC-Trans monitoring system's tracking of student engagement and the results of the quizzes and expert assessments, and thus, it can be assumed that the monitoring system is capable of tracking student engagement in a classroom setting.

Furthermore, the transitions in teaching phases and the corresponding changes in engagement levels are represented by the divided regions in Fig. 3, where:

Region① Interactive Teaching: the teacher selects students to answer questions based on pre-class preparation exercises;

Region② Lecture-based Teaching: the teacher explains Knowledge Point 1;

Region③ Interactive Teaching: the teacher writes Knowledge Point 2 on the board and selects students to answer questions;

Region④ Lecture-based Teaching: the teacher explains Knowledge Point 3;

Region⑤ Interactive Teaching: students complete written exercises on Knowledge Point 3, while the teacher moves around the classroom to provide individual assistance;

Region⑥ Interactive Teaching: the teacher writes Knowledge Point 4 on the board and selects students to answer questions.

Through the relationship between the teaching phases and student engagement, we can find that in lecture-based teaching phases, students' average engagement gradually decreases over time. In interactive teaching phases, however, students' average engagement tends to show a noticeable increase.

E. Result and Discussion

The MSC-Trans network is trained on NVIDIA RTX4080 GPU and it takes 609 ms on average to detect students' engagement per frame.

Using each knowledge point's corresponding time period and its average engagement level as the baseline, continuous monitoring of student engagement in videos is conducted within a 40-min lesson, calculating the average engagement level per minute in the detection results. After the lesson, a quiz on each knowledge point is conducted, and the tested knowledge points are mapped to their corresponding classroom teaching periods. In addition, expert teachers also rate the student engagement for the time periods associated with the classroom knowledge

TABLE III
MSE OF MTL-CLASSIFIER

Feature \ Compare Range	Whole Engagement Statement	Lower Engagement Statement	Higher Engagement Statement
Eyes	1.51	1.18	1.49
Head Gesture	3.17	1.87	4.48
Writing	2.85	2.88	2.06
Using Phone	1.96	1.4	2.64

TABLE IV
FRÉCHET SIMILARITY OF SERIES FUSION PREDICTION

Feature Group \ Compare Range	Whole Engagement Statement	Lower Engagement Statement	Higher Engagement Statement
Feature Group-All	0.852	0.919	0.818
Feature Group 1 (Eyes + Head + Phone)	0.561	0.731	0.472
Feature Group 2 (Eyes + Head + Writing)	0.591	0.335	0.695
Feature Group 3 (Head + Writing + Phone)	0.502	0.743	0.524

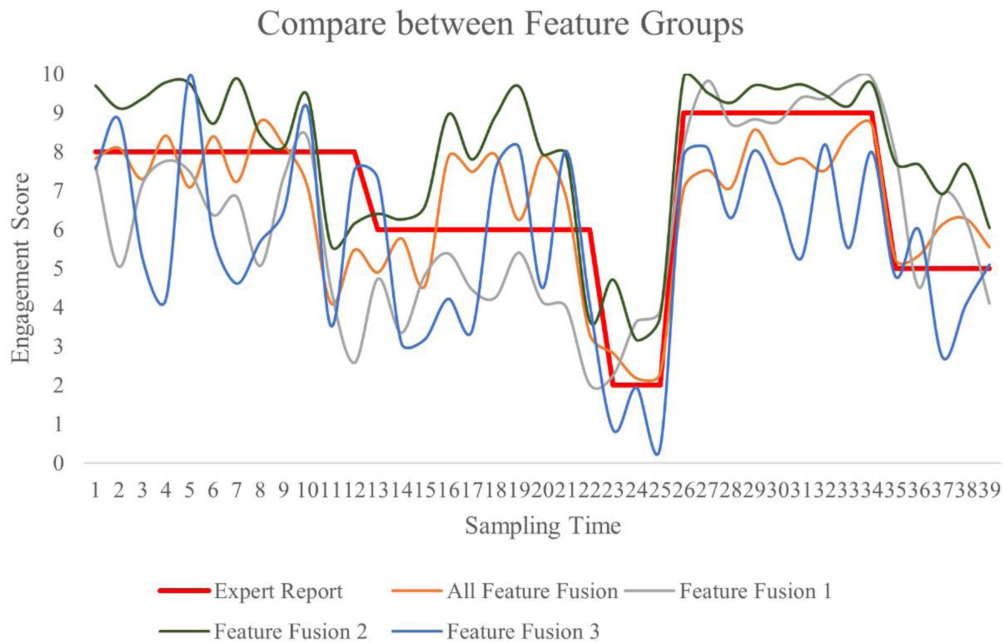


Fig. 6. Feature groups compare.

points. The correspondence among these three aspects is shown in Fig. 4.

The MTL Classifier assigns probability scores to four behavioral characteristics in student classroom behavior images: eye state, head posture, writing status, and mobile phone usage. The detection results are shown in Fig. 5.

The comparison of its MSE with the true values is given in Table III.

The MSC-Trans network can track engagement by different combinations of features that can be arbitrarily reconfigured to verify this property and also to gain a deeper understanding of the interactions between features; four groups of experiments with different feature combinations are arranged to comprehend the interplay more thoroughly between each feature dimension. The results of the combined feature experiments are given in Table IV and Fig. 6.

TABLE V
ENGAGEMENT DETECTION AND TRACKING ACCURACY ON THE DAISEE DATASET

	Dataset: DAiSEE	Average Accuracy%	Fréchet Similarity (10 s)%
Classifier	InceptionNet	47.11	67.32
	ResNet50+LSTM	58.84	76.28
	DCNN+LSTM	63.33	77.64
	Resnet18+TCN	63.90	80.03
	MSC-Trans	61.86	85.21

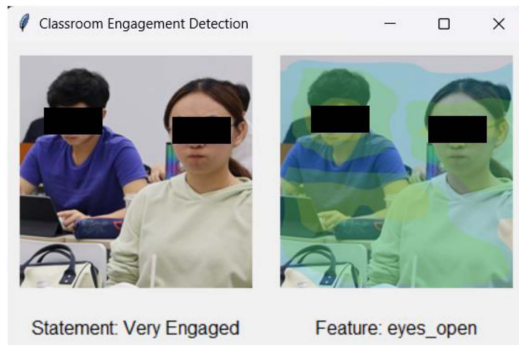


Fig. 7. Teacher UI.

To improve the interpretability of the network's results for educators, we designed an educator-oriented UI (see Fig. 7, where the left image displays the real-time detection of students' learning states in class as determined by the MSC-Trans network. The right image highlights the key features influencing student engagement at that moment, with different color regions indicating their distribution—warm tones represent high-weight feature areas, while cool tones indicate lower-weight areas—with an integrated display of the results of student classroom engagement tests and the distribution of associated features. This UI can make the use of the network more convenient for educators and enable them to have a deeper understanding of the students' classroom engagement status and related features.

The network model exhibits the highest tracking accuracy when all features are combined, demonstrating a good ability to reflect the trend of student engagement in the classroom over time.

F. Comparison With State of the Art

To demonstrate the effectiveness of the MTL Classifier and sequence fusion encoder in MSC-Trans Engagement Detecting Network, several tests are proceeded on the DAiSEE dataset for the image detection accuracy (MSE) and sequence tracking accuracy (Fréchet Similarity) of various common convolutional image + temporal detection methods. The results are given in Table V.

It can be observed that MSC-Trans Engagement Detecting Network achieves a commendable level of accuracy in single image detection, but because of the delay caused by the attention module in Transformer structure used in multilabel classification and temporal feature fusion, there is a slight gap between the average accuracy and the state-of-the-art networks. But in

the task of engagement tracking, using Fréchet Similarity as the standard, it can reach an optimal tracking accuracy of 85.21% in the task of temporal tracking of student engagement states. This indicates that the network structure has advantages in tracking the changes in student classroom engagement over time. Moreover, the MSC-Trans network maintains a high level of accuracy and tracking precision, while still enabling tracking prediction of inputs with different combinations of features, it has a high degree of flexibility in real classroom environments.

V. CONCLUSION AND FUTURE WORK

In this study, MSC-Trans Engagement Detecting Network, a video-based network for detecting student classroom engagement, is constructed, which is capable of performing high-accuracy, effective, and continuous tracking of students' classroom states. The main features of the network are shown as follows.

- 1) Multilabel image prediction network: This network uses the embedding attention structure to improve its logical reasoning about the associated action features, which can help increase the accuracy of logistic reasoning when analyzing learners' classroom videos, sequentially outputting the probability scores of various highly resolute behavioral features appearing in the frames.
- 2) Expandable sequence fusion prediction network: This network is capable of predicting subsequent items based on the combination of multidimensional feature sequences. This part of the network can output a predictive curve of the learner's learning traits over time. In addition, the temporal attention structure allows the researcher to adjust the combination of features as needed, making the prediction task more flexible.

To test the effectiveness of this fusion network, the classroom state of learners is firstly recorded with cameras, ensuring a strict temporal response relationship of learners' behaviors, and uses the recorded image data to train the multilabel image prediction network. Next, key features of the learner's classroom behavior and emotional expressions are extracted and quantified using the multilabel image prediction network, allowing for real-time recording and tracking of individual learners and overall classroom status. This is followed by generating a list of learners' classroom behavior and state trajectories over time to train the expandable sequence fusion prediction network. The sequence fusion network is utilized for the fusion training of the learners' temporal features to obtain the learners' learning state curve,

achieving real-time prediction and long-term tracking of learner engagement.

Experimental results indicate that our network has an excellent performance, reaching 85.21% Fréchet Similarity in the real-time tracking task of student classroom engagement, which shows a significant consistency with the standard engagement curve recorded by experts, reflecting the changing trends of student engagement over time well and demonstrating the reflective capability of different behavioral features on changes in engagement.

After validating the accuracy of the engagement monitoring network, we conducted a teaching experiment using the network in a real classroom environment. During the experiment, teachers were able to identify teaching segments where student engagement was low through the monitoring results, enabling them to adjust their teaching strategies, such as incorporating more interactive activities or simplifying the content. In addition, the feature heatmaps generated by the monitoring system provided teachers with a visual understanding of the relationship between students' facial expressions, behaviors, and engagement levels. This helped teachers better recognize students' participation states, optimize their teaching methods, and ultimately improve classroom quality and student learning outcomes.

Although the monitoring network demonstrated strong performance in the experiment, it also revealed several limitations. The network's structure is relatively complex, with frequent data transmissions, and its training and inference processes can only run efficiently on high-end GPU-equipped devices. Consequently, the network exhibits weaker compatibility with commonly used smart education platform systems, posing certain challenges for large-scale classroom deployment, especially in educational institutions with limited hardware resources. To address these challenges, we will continue to optimize the network structure to reduce model parameter size and computational requirements, as well as refine data interfaces and user configurations to ensure compatibility with low-performance devices and existing smart education systems. In addition, lightweight models will be developed using distillation methods to make the network compatible with low-performance devices. Furthermore, constructing a dedicated training dataset tailored to the common complex features observed in real classroom scenarios will be considered, enabling the network to achieve improved generalization capabilities.

In the future, integrating a student engagement detection system with real teaching scenarios can help teachers better understand students' participation, enabling them to adjust teaching strategies and improve classroom effectiveness. Teachers can use the system to monitor student engagement in real-time, taking actions such as introducing interactive activities when engagement drops to increase participation. After class, teachers can also use the engagement detection records to design teaching content and optimize course design.

In addition, this network could further be expanded to incorporate features, such as teacher engagement and teacher-student interaction behaviors, ultimately enabling continuous monitoring of the overall classroom teaching atmosphere. This would provide more persuasive data to support classroom optimization and offer effective assistance for the collaborative progress of

both students and teachers. With the introduction of more feature selection techniques, advanced anti-interference measures can further be adopted to enhance the system's robustness in real-world environments. First, more powerful image completion techniques, such as image restoration methods based on generative adversarial networks, can be employed to address image occlusion issues in complex scenarios. In addition, integrating the latest attention mechanism modules into the network structure, such as dynamic adaptive attention or multimodal focus mechanisms, can improve the system's ability to capture critical student behaviors. Reinforcement learning is another potential approach, where continuous classroom scenario simulation training could further enhance the network's robustness in disruptive environments. Through the integration and optimization of these technologies, future engagement monitoring systems are expected to achieve new levels of performance and applicability, becoming a vital tool for supporting classroom teaching.

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